2.3 Laws of illumination

The illumination on a surface depends upon the luminous intensity, distance between the source and surface and the direction of rays of light. It is governed by following laws.

1. Inverse square law
2. Lambert's cosine law

2.3.1 Inverse square law

It states that the illumination of a surface is inversely proportional to the square of the distance of the surface from the source.

$$E \alpha \frac{1}{d^2}$$

2.3.2 Lambert's cosine law

This law states that the illumination on any surface is proportional to the cosine of angle between the directions of the incident flux and perpendicular to the area.

$$E = \frac{1}{d^2} \cos \theta$$

2.3.3 Light

The radiant energy from a hot body which produces the visual sensation on human eye is called light.

2.3.4 Electrical method of producing light

Following are the methods of producing light:

1. Developing arc between two electrodes.
2. Passing a current through a filament.
3. Electric discharge through vapours or gases.

2.4 Arc lamp

The principle of an arc lamp is that when two electrodes carrying current are separated through a small distance, an arc is struck between them. The arc lamps were used in the past for street lighting purposes but nowadays these are used when extreme brightness is required. Carbon arc lamp is most commonly used arc lamp.

2.4.1 Carbon arc lamp

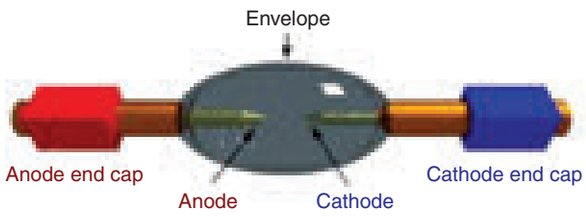


Fig. 2.3 Arc lamp

Arc lamp is shown in figure 2.3. Carbon arc lamp is the oldest type of lamp and is still being implied in cinema projectors and searchlights. It consists of two hard carbon rods (Electrodes). The diameter of positive electrode is double to that of negative electrode. The negative electrode is generally fixed and positive electrode is placed in adjustable holder and the process is manual or automatic. The arc consists of carbon vapours surrounded by orange red zone of burning carbon and pale green flames.

When the lamp is switched OFF, the two electrodes touch each other due to spring pressure on positive electrode. When the supply is ON a large current flows through electrodes. The temperature of carbon electrode is increased and thus

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